

Lógica y Teoría de Números

Conjuntos

Representar en un diagrama de Venn

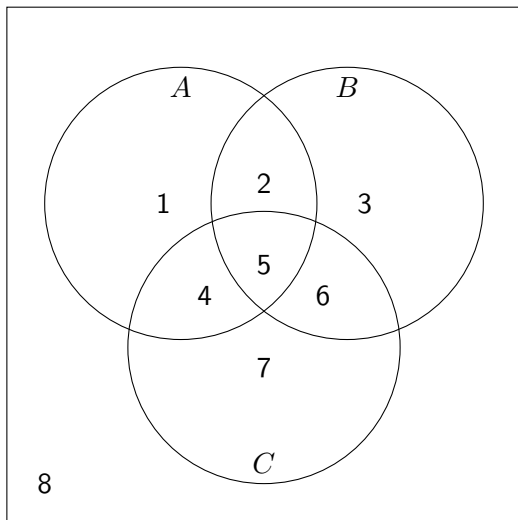
Representar en un Diagrama de Venn

Ejercicio

Sean A , B y C subconjuntos del conjunto universal $\mathcal{U}$. Representar el conjunto $A \cup (B^c \cap C^c)$ en un diagrama de Venn.

Para resolverlo primero vamos a enumerar cada sector del diagrama de Venn. Luego, para cada operación de conjuntos, escribimos qué partes "sombrearíamos" para representarla.

Enumeramos cada sector del diagrama de Venn.



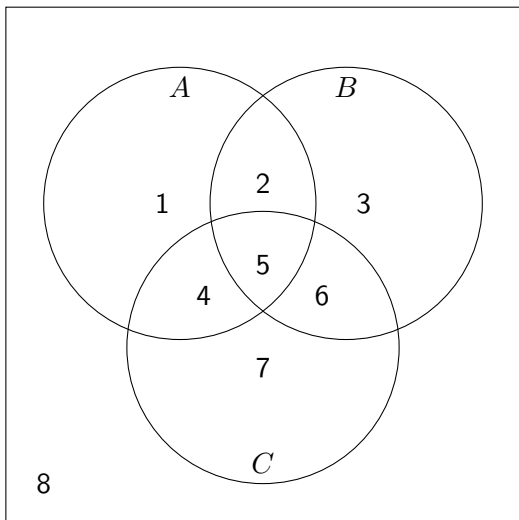
$$B^c =$$

$$A =$$

$$C^c =$$

$$A \cup (B^c \cap C^c) =$$

$$B^c \cap C^c =$$



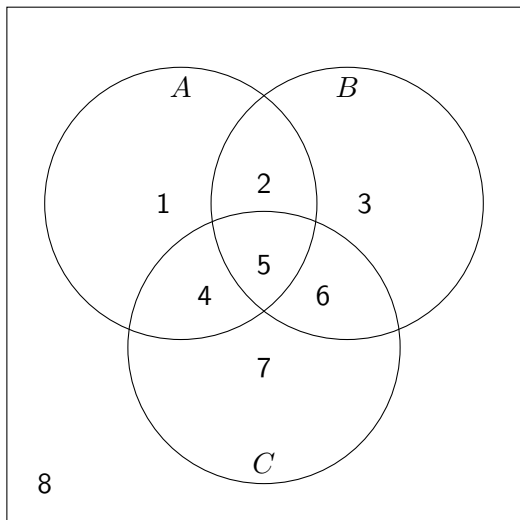
$$B^c = 1 + 4 + 7 + 8$$

$$C^c =$$

$$B^c \cap C^c =$$

$$A =$$

$$A \cup (B^c \cap C^c) =$$



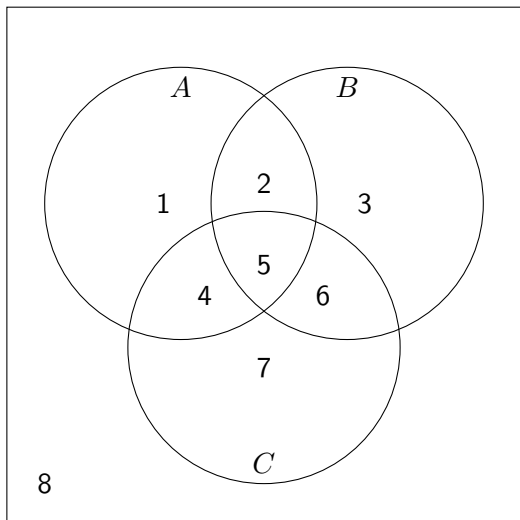
$$B^c = 1 + 4 + 7 + 8$$

$$C^c = 1 + 2 + 3 + 8$$

$$B^c \cap C^c =$$

$$A =$$

$$A \cup (B^c \cap C^c) =$$



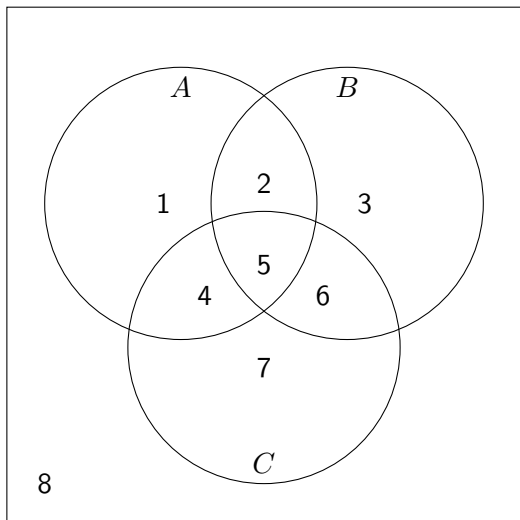
$$B^c = 1 + 4 + 7 + 8$$

$$C^c = 1 + 2 + 3 + 8$$

$$B^c \cap C^c = 1 + 8$$

$$A =$$

$$A \cup (B^c \cap C^c) =$$



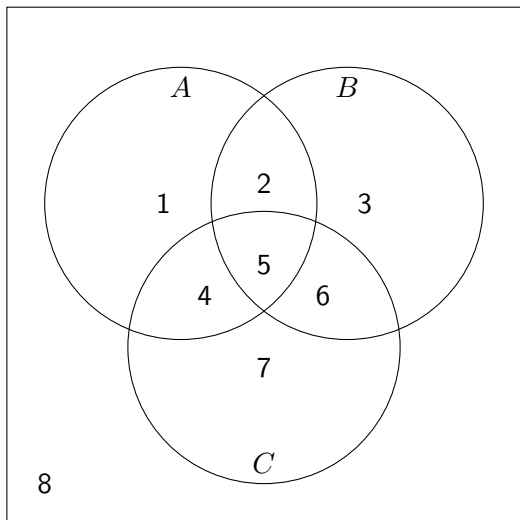
$$B^c = 1 + 4 + 7 + 8$$

$$C^c = 1 + 2 + 3 + 8$$

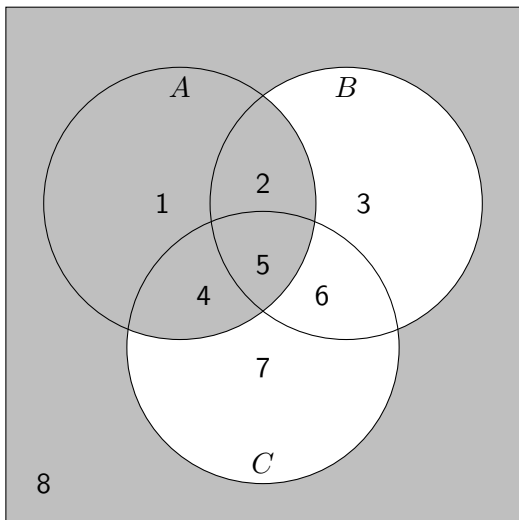
$$B^c \cap C^c = 1 + 8$$

$$A = 1 + 2 + 4 + 5$$

$$A \cup (B^c \cap C^c) =$$



$$\begin{aligned} B^c &= 1 + 4 + 7 + 8 & C^c &= 1 + 2 + 3 + 8 & B^c \cap C^c &= 1 + 8 \\ A &= 1 + 2 + 4 + 5 & A \cup (B^c \cap C^c) &= 1 + 2 + 4 + 5 + 8 \end{aligned}$$



$$B^c = 1 + 4 + 7 + 8$$

$$C^c = 1 + 2 + 3 + 8$$

$$B^c \cap C^c = 1 + 8$$

$$A = 1 + 2 + 4 + 5$$

$$A \cup (B^c \cap C^c) = 1 + 2 + 4 + 5 + 8$$